



Fuhua-Parallel Sec 3 Express EOY 2026 Paper 1 — Marking Scheme Worked Solutions + Marking Scheme

Cambridge Learning Group · Fuhua-parallel practice paper · Paper 1 · 2026
14 pages · 80 marks

How to read this booklet:

BLACK — the question, written ON the original question paper

BLUE — the full step-by-step working and the reason for each line

RED — the marking scheme: where each B1 / M1 / A1 is earned, the traps,
and the key concept the question is testing

Mark codes: B1 independent answer mark · M1 method mark (earned even if

the answer is wrong) · A1 accuracy mark (only if the method mark was earned).

Answer all the questions.

- 1 Find the range of values of k for which the curve $y = x^2 + (k-2)x + 4$ lies entirely above the line $y = 2x - k$. [4]

Curve above line for all x :

$$x^2 + (k-2)x + 4 > 2x - k \text{ for all } x$$

$$x^2 + (k-4)x + (4+k) > 0 \text{ for all } x$$

A positive quadratic is always above the x -axis when it has no real root:

$$(k-4)^2 - 4(4+k) < 0$$

$$k^2 - 8k + 16 - 16 - 4k < 0$$

$$k^2 - 12k < 0$$

$$k(k-12) < 0$$

$$\therefore 0 < k < 12$$

Key concept: above for ALL x means the difference is a positive quadratic with no real root.

M1 - forms the difference and gets

$$x^2 + (k-4)x + (4+k) > 0$$

M1 - uses discriminant < 0 (not ≤ 0 , and not $0 < \text{discriminant}$)

$$\text{M1 - reduces to } k^2 - 12k < 0$$

$$\text{A1 - } 0 < k < 12$$

Trap - $k < 0$ or $k > 12$ comes from solving

$$k(k-12) > 0; \text{ sketch the parabola in } k.$$

(a) Express $\frac{5-2\sqrt{3}}{2\sqrt{3}-3}$ in the form $a+b\sqrt{3}$, where a and b are rational numbers. [3]

$$\frac{5-2\sqrt{3}}{2\sqrt{3}-3} \times \frac{2\sqrt{3}+3}{2\sqrt{3}+3}$$

$$\text{denominator} = (2\sqrt{3})^2 - 3^2 = 12 - 9 = 3$$

$$\text{numerator} = 10\sqrt{3} + 15 - 4(3) - 6\sqrt{3}$$

$$\text{numerator} = 4\sqrt{3} + 3$$

$$\frac{3+4\sqrt{3}}{3} = 1 + \frac{4}{3}\sqrt{3}$$

$$\therefore a=1, b=\frac{4}{3}$$

(b) Solve the equation $\sqrt{3x+7} - \sqrt{x+1} = 2$. [3]

$$\sqrt{3x+7} = 2 + \sqrt{x+1}$$

$$3x+7 = 4 + 4\sqrt{x+1} + x+1$$

$$2x+2 = 4\sqrt{x+1}$$

$$x+1 = 2\sqrt{x+1}$$

$$\text{let } w = \sqrt{x+1}: w^2 = 2w$$

$$w(w-2) = 0, \text{ so } w=0 \text{ or } w=2$$

$$x+1=0 \text{ or } x+1=4$$

$$\text{check } x=-1: \sqrt{4} - \sqrt{0} = 2 \quad \checkmark$$

$$\text{check } x=3: \sqrt{16} - \sqrt{4} = 4 - 2 = 2 \quad \checkmark$$

$$\therefore x=-1 \text{ or } x=3$$

Key concept: the conjugate of $2\sqrt{3}-3$ is $2\sqrt{3}+3$.

M1 - multiplies by the conjugate $2\sqrt{3}+3$

M1 - denominator $12-9=3$

A1 - $a=1$ and $b=\frac{4}{3}$

Note - the question says RATIONAL, not integer — $\frac{4}{3}$ is acceptable.

Key concept: isolate one surd, square, then isolate the surd that remains.

M1 - squares after isolating one surd

M1 - reaches $x+1=2\sqrt{x+1}$ and squares again

A1 - BOTH $x=-1$ and $x=3$

Trap - both roots satisfy the original equation — neither may be rejected. Discarding one loses the A1.

- 3 It is given that $f(x) = 2x^3 + ax^2 + bx - 6$, where a and b are constants. $(x-2)$ is a factor of $f(x)$, and $f(x)$ leaves a remainder of -12 when divided by $(x+1)$.

(a) Find the value of a and of b .

[4]

$$(x-2) \text{ a factor} \Rightarrow f(2) = 0:$$

$$16 + 4a + 2b - 6 = 0$$

$$2a + b = -5 \quad \dots(1)$$

$$f(-1) = -12: -2 + a - b - 6 = -12$$

$$a - b = -4 \quad \dots(2)$$

$$(1) + (2): 3a = -9, \text{ so } a = -3$$

$$\text{from (2): } b = a + 4 = 1$$

$$\therefore a = -3, b = 1$$

$$\text{M1 - } f(2) = 0 \text{ used}$$

$$\text{A1 - } 2a + b = -5$$

$$\text{M1 - } f(-1) = -12 \text{ used}$$

$$\text{A1 - } a = -3 \text{ and } b = 1$$

$$\text{Trap - remainder } -12 \text{ means } f(-1) = -12, \text{ not } f(-1) = 0.$$

(b) Solve the equation $f(x) = 0$, explaining why it has only one real root.

[3]

$$f(x) = 2x^3 - 3x^2 + x - 6$$

$$f(x) = (x-2)(2x^2 + x + 3)$$

$$x - 2 = 0 \text{ gives } x = 2$$

$$\text{for } 2x^2 + x + 3 = 0:$$

$$\text{discriminant} = 1^2 - 4(2)(3) = 1 - 24 = -23 <$$

$$\text{so the quadratic factor has no real root}$$

$$\therefore x = 2 \text{ is the only real root}$$

Key concept: a cubic with one real root has an irreducible quadratic factor.

$$\text{M1 - divides out } (x-2) \text{ to get } 2x^2 + x + 3$$

$$\text{A1 - discriminant } -23 < 0 \text{ computed}$$

A1 - states $x = 2$ is the only real root, because the quadratic has no real root

Note - the explanation is required — quoting $x = 2$ alone scores at most M1 A1.

- (i) Explain why $\frac{2x^4+2x^3+6x^2+4x+4}{x^2(x^2+4)}$ cannot be expressed in the form [2]

$$\frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+4}, \text{ where } A, B, C \text{ and } D \text{ are constants.}$$

Numerator degree = 4; denominator

$x^2(x^2+4)$ has degree 4.

The fraction is improper (degree of top $\geq$ degree of bottom).

The given form is a sum of proper fractions, so it can only represent a proper fraction.

A constant term must be included first, by division.

B1 - identifies both degrees as 4, so the fraction is improper

B1 - states that a constant (the quotient) is needed before the proper fractions

Note - saying only "the powers are the same" without the conclusion scores B1.

- (ii) Express $\frac{2x^4+2x^3+6x^2+4x+4}{x^2(x^2+4)}$ in partial fractions. [6]

$$\frac{2x^4+2x^3+6x^2+4x+4}{x^4+4x^2} = 2 + \frac{2x^3-2x^2+4x+4}{x^2(x^2+4)}$$

$$\frac{2x^3-2x^2+4x+4}{x^2(x^2+4)} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+4}$$

$$2x^3-2x^2+4x+4 = Ax(x^2+4) + B(x^2+4) + (Cx+D)x^2$$

$$x=0: 4=4B, \text{ so } B=1$$

$$x^3: 2=A+C; \quad x^2: -2=B+D, \text{ so } D=-3$$

$$x: 4=4A, \text{ so } A=1, \text{ hence } C=1$$

$$\therefore 2 + \frac{1}{x} + \frac{1}{x^2} + \frac{x-3}{x^2+4}$$

Key concept: divide first, then split; x^2 needs TWO terms and x^2+4 needs $Cx+D$.

M1 - divides to get the quotient 2

M1 - correct form with $\frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+4}$

M1 - clears denominators and substitutes or compares

A1 - $B=1$ and $A=1$

A1 - $C=1$

A1 - $D=-3$

Note - check at $x=1$: LHS = $\frac{18}{5}$,

$$\text{RHS} = 2 + 1 + 1 - \frac{2}{5} = \frac{18}{5} \checkmark$$

(a) Solve the equation $3^{2x+1} - 10(3^x) + 3 = 0$. [3]

$$3^{2x+1} = 3(3^x)^2$$

$$\text{let } u = 3^x: 3u^2 - 10u + 3 = 0$$

$$(3u - 1)(u - 3) = 0$$

$$u = \frac{1}{3} \text{ or } u = 3$$

$$3^x = 3^{-1} \text{ or } 3^x = 3^1$$

$$\therefore x = -1 \text{ or } x = 1$$

Key concept: $3^{2x+1} = 3 \cdot (3^x)^2$ turns it into a quadratic.

M1 - writes 3^{2x+1} as $3(3^x)^2$ and substitutes $u = 3^x$

M1 - solves the quadratic in u

A1 - $x = -1$ and $x = 1$

Trap - $3^{2x+1} \neq (3^x)^2$ - the extra factor of 3 is what makes it work.

(b) Solve the simultaneous equations [4]

$$\log_2 x + \log_2 y = 5, \log_4 x - \log_4 y = \frac{1}{2}$$

$$\log_2 x + \log_2 y = 5 \Rightarrow \log_2(xy) = 5$$

$$xy = 32 \dots (1)$$

$$\log_4 x - \log_4 y = \frac{1}{2} \Rightarrow \log_4 \frac{x}{y} = \frac{1}{2}$$

$$\frac{x}{y} = 4^{\frac{1}{2}} = 2, \text{ so } x = 2y \dots (2)$$

$$(2) \text{ into } (1): 2y^2 = 32$$

$$y^2 = 16, \text{ and } y > 0, \text{ so } y = 4$$

$$\therefore x = 8, y = 4$$

Key concept: combine each equation into a single log, then remove the log.

M1 - $\log_2(xy) = 5$ giving $xy = 32$

M1 - $\log_4 \frac{x}{y} = \frac{1}{2}$ giving $\frac{x}{y} = 2$

A1 - $y = 4$

A1 - $x = 8$

Trap - $y = -4$ must be rejected: $\log_2 y$ requires $y > 0$.

- 6 The mass, M grams, of a culture of bacteria after t hours is modelled by $M = M_0 e^{kt}$, where M_0 and k are constants. The mass is 8 g when $t = 1$ and 27 g when $t = 4$.

(a) Show that $k = \ln \frac{3}{2}$ and find the exact value of M_0 . [4]

$$t = 1: 8 = M_0 e^k; \quad t = 4: 27 = M_0 e^{4k}$$

$$\text{dividing: } \frac{27}{8} = e^{3k}$$

$$3k = \ln \frac{27}{8} = 3 \ln \frac{3}{2}$$

$$\therefore k = \ln \frac{3}{2} \quad (\text{shown})$$

$$e^k = \frac{3}{2}, \text{ so } 8 = M_0 \left(\frac{3}{2} \right)$$

$$\therefore M_0 = \frac{16}{3} \text{ g}$$

(b) Find the least number of complete hours after which the mass first exceeds 200 g.

$$\frac{16}{3} \left(\frac{3}{2} \right)^t > 200$$

$$\left(\frac{3}{2} \right)^t > \frac{600}{16} = 37.5$$

$$t \ln \frac{3}{2} > \ln 37.5$$

$$t > \frac{3.6243}{0.40546} = 8.938 \dots$$

$\therefore$ the least number of complete hours is 9

Key concept: divide the two equations to eliminate M_0 .

M1 - both equations written and divided

$$\text{M1 - } e^{3k} = \frac{27}{8}$$

$$\text{A1 - } k = \ln \frac{3}{2} \text{ shown, using } \ln \frac{27}{8} = 3 \ln \frac{3}{2}$$

$$\text{A1 - } M_0 = \frac{16}{3}$$

Note - exact value required — 5.33 alone loses the A1.

M1 - forms the inequality and isolates $\left(\frac{3}{2} \right)^t$ [3]

M1 - takes logs correctly

A1 - 9 hours

Trap - $t = 8.94$ rounds to 8.9, but the mass has not yet exceeded 200 g at $t = 8$ — the answer is the next whole hour, 9.

- (a) In the expansion of $(1+2x)^n$, the coefficient of x^2 is 96 more than the coefficient of x . Find the value of n . [3]

coefficient of x : $\binom{n}{1}2 = 2n$

coefficient of x^2 : $\binom{n}{2}2^2 = 2n(n-1)$

$$2n(n-1) = 2n + 96$$

$$2n^2 - 4n - 96 = 0$$

$$n^2 - 2n - 48 = 0$$

$$(n-8)(n+6) = 0$$

$n > 0$, so $\therefore n = 8$

M1 - both coefficients written in terms of n

M1 - forms $2n(n-1) = 2n + 96$ and reduces to a quadratic

A1 - $n = 8$, rejecting $n = -6$

Trap - $\binom{n}{2}$ must carry the factor 2^2 from $(2x)^2$.

- (b) Using this value of n , find the coefficient of x^5 in the expansion of $(2-x)(1+2x)^n$. [4]

$(1+2x)^8$: coefficient of

$$x^5 = \binom{8}{5}2^5 = 56(32) = 1792$$

coefficient of $x^4 = \binom{8}{4}2^4 = 70(16) = 1120$

In $(2-x)(1+2x)^8$ the x^5 term comes from

$2 \times (x^5 \text{ term})$ and $(-x) \times (x^4 \text{ term})$

$$\text{coefficient} = 2(1792) - 1(1120)$$

$$\text{coefficient} = 3584 - 1120$$

$\therefore$ coefficient of $x^5 = 2464$

Key concept: with a bracket in front, TWO terms of the expansion contribute.

M1 - coefficient of x^5 in $(1+2x)^8$ as $\binom{8}{5}2^5$

M1 - coefficient of x^4 as $\binom{8}{4}2^4$

M1 - combines as $2 \times 1792 - 1 \times 1120$

A1 - 2464

Trap - forgetting the x^4 contribution gives 3584; the sign of $-x$ matters.

8 The points $A(-1, 2)$ and $B(5, 4)$ lie on a circle C_1 , and AB is a diameter of C_1 .

(a) Find the equation of C_1 . [3]

AB a diameter, so the centre is the midpoint of AB :

$$\text{centre} = \left(\frac{-1+5}{2}, \frac{2+4}{2} \right) = (2, 3)$$

$$AB^2 = (5+1)^2 + (4-2)^2 = 36+4=40$$

$$\text{radius} = \frac{\sqrt{40}}{2}, \text{ so } r^2 = 10$$

$$\therefore (x-2)^2 + (y-3)^2 = 10$$

(b) The tangent to C_1 at A meets the y -axis at P . Find the coordinates of P . [3]

$$\text{gradient of radius } CA = \frac{2-3}{-1-2} = \frac{1}{3}$$

tangent $\perp$ radius, so gradient of

$$\text{tangent} = -3$$

$$\text{at } A(-1, 2): y-2 = -3(x+1)$$

$$y = -3x - 1$$

$$\text{at the } y\text{-axis, } x=0, \text{ so } y = -1$$

$$\therefore P(0, -1)$$

M1 - centre as the midpoint of the diameter

$$\text{M1} - r^2 = 10 \text{ from } \left(\frac{AB}{2} \right)^2 \text{ or from centre to } A$$

$$\text{A1} - (x-2)^2 + (y-3)^2 = 10$$

Key concept: the tangent at a point is perpendicular to the radius there.

$$\text{M1} - \text{gradient of the radius} = \frac{1}{3}$$

M1 - uses $m_1 m_2 = -1$ to get -3 and forms the tangent

$$\text{A1} - P(0, -1)$$

(c) Show that P lies outside C_1 , and find the exact length of the tangent from P to C_1 . [3]

distance from $P(0, -1)$ to the centre $(2, 3)$:

$$d^2 = (2 - 0)^2 + (3 + 1)^2 = 4 + 16 = 20$$

$$d^2 = 20 > 10 = r^2, \text{ so } P \text{ lies outside } C_1$$

$$\text{tangent length} = \sqrt{d^2 - r^2}$$

$$\text{tangent length} = \sqrt{20 - 10}$$

$$\therefore \text{tangent length} = \sqrt{10}$$

Key concept: the tangent, the radius and the line to the centre form a right triangle.

M1 - $d^2 = 20$ computed

A1 - compares $d^2 = 20$ with $r^2 = 10$ and concludes P is outside

A1 - $\sqrt{10}$, exact

Trap - comparing d with r^2 , or leaving $\sqrt{10}$ as 3.16 when EXACT is asked for.

(d) Find the area of triangle PAB . [3]

$$P(0, -1), A(-1, 2), B(5, 4)$$

$$\text{area} = \frac{1}{2} |x_P(y_A - y_B) + x_A(y_B - y_P) + x_B(y_P - y_A)|$$

$$\text{area} = \frac{1}{2} |0(2 - 4) + (-1)(4 + 1) + 5(-1 - 2)|$$

$$\text{area} = \frac{1}{2} |0 - 5 - 15|$$

$$\text{area} = \frac{1}{2} (20)$$

$$\therefore \text{area} = 10 \text{ units}^2$$

M1 - a correct area method (shoelace, or $\frac{1}{2} \times \text{base} \times \text{height}$)

M1 - correct substitution of all three points

A1 - 10 units²

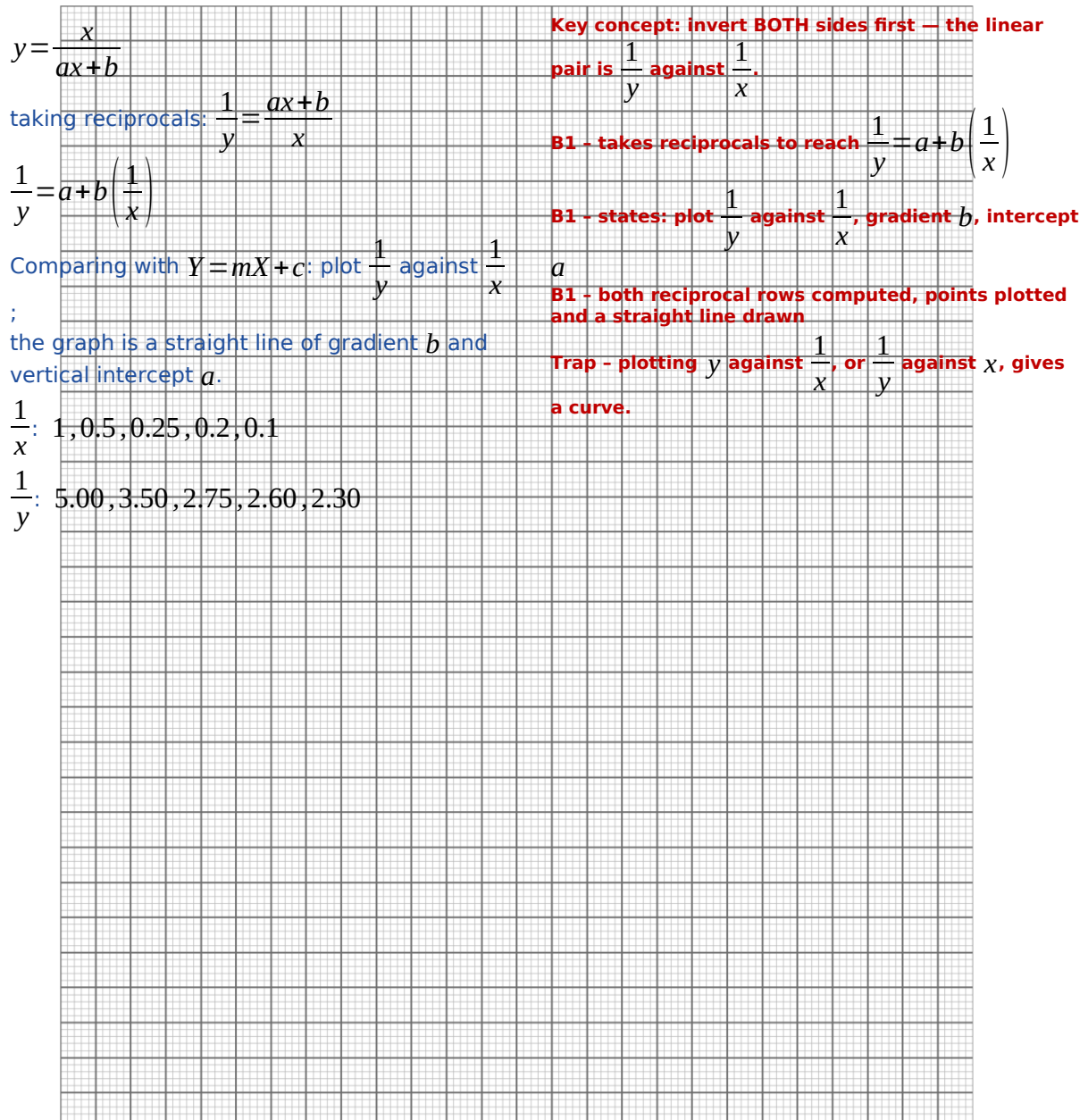
Note - $PA \perp AB$ (the tangent), so

$$\frac{1}{2} (PA)(AB) = \frac{1}{2} \sqrt{10} \sqrt{40} = 10 \text{ is a neat check.}$$

- 9 Two variables x and y are related by the equation $y = \frac{x}{ax+b}$, where a and b are constants. The table below shows corresponding values of x and y .

x	1	2	4	5	10
y	0.200	0.286	0.364	0.385	0.435

- (a) Explain how a straight line graph may be drawn to represent this equation, and draw it for the given data on the grid below. [3]



(b) Use your graph to estimate the value of a and of b .

[3]

$$\text{gradient } b = \frac{5.00 - 2.30}{1.0 - 0.1} = \frac{2.70}{0.9}$$

$$b \approx 3$$

$$\text{intercept: } \frac{1}{y} = a \text{ when } \frac{1}{x} = 0$$

$$a = 5.00 - 3(1.0)$$

$$\therefore a \approx 2, b \approx 3$$

Key concept: read two points off the **DRAWN LINE**, as far apart as it allows.

M1 - gradient from two points on the line

A1 - $b \approx 3$

M1 - intercept read at $\frac{1}{x} = 0$, or substitution into the line

A1 - $a \approx 2$

Note - accept a from 1.8 to 2.2 and b from 2.8 to 3.2.

Note - check: $y = \frac{x}{2x+3}$ gives $y = 0.364$ at $x = 4$

✓

- 10 Given that A is obtuse and $\cos A = -p$, where $0 < p < 1$, express each of the following in terms of p .

(i) $\sin A$
 A obtuse, so A is in quadrant 2: $\sin A > 0$
 $\sin^2 A = 1 - \cos^2 A = 1 - (-p)^2 = 1 - p^2$
 $\therefore \sin A = \sqrt{1 - p^2}$

[2]
M1 - uses $\sin^2 A + \cos^2 A = 1$
A1 - $\sqrt{1 - p^2}$, positive because A is obtuse
Trap - $\pm \sqrt{1 - p^2}$ loses the A1 — the quadrant decides the sign.

(ii) $\tan(180^\circ - A)$
 $\tan(180^\circ - A) = -\tan A$
 $\tan A = \frac{\sin A}{\cos A} = \frac{\sqrt{1 - p^2}}{-p}$
 $\therefore \tan(180^\circ - A) = \frac{\sqrt{1 - p^2}}{p}$

[2]
M1 - $\tan(180^\circ - A) = -\tan A$
A1 - $\frac{\sqrt{1 - p^2}}{p}$
Note - two negatives cancel: the answer is positive.

(iii) $\operatorname{cosec}(90^\circ + A)$
 $\sin(90^\circ + A) = \cos A = -p$
 $\operatorname{cosec}(90^\circ + A) = \frac{1}{\sin(90^\circ + A)}$
 $\therefore \operatorname{cosec}(90^\circ + A) = -\frac{1}{p}$

[2]
M1 - $\sin(90^\circ + A) = \cos A$
A1 - $-\frac{1}{p}$
Trap - $90^\circ + A$, not $90^\circ - A$: the sine of $90^\circ + A$ is $+\cos A$, and $\cos A$ is negative here.

- 11 The function f is defined by $f(x) = a - b \cos \frac{x}{2}$ for $0 \leq x \leq 4\pi$, where a and b are positive constants. The maximum value of f is 7 and the minimum value of f is -1 .

(a) Find the value of a and of b .

[3]

$-\cos \frac{x}{2}$ is greatest when $\cos \frac{x}{2} = -1$, so

$$\text{maximum} = a + b = 7$$

least when $\cos \frac{x}{2} = 1$, so

$$\text{minimum} = a - b = -1$$

adding: $2a = 6$, so $a = 3$

$$\therefore a = 3, b = 4$$

(b) State the period of f and sketch the graph of $y = f(x)$ for $0 \leq x \leq 4\pi$ on the axes below.

[3]

period = $\frac{2\pi}{\frac{1}{2}} = 4\pi$, so exactly one cycle on

$$0 \leq x \leq 4\pi$$

$$f(0) = 3 - 4(1) = -1 \text{ (minimum at } x = 0)$$

$$f(2\pi) = 3 - 4(-1) = 7 \text{ (maximum at } x = 2\pi)$$

$$f(4\pi) = -1$$

curve crosses $y = 3$ at $x = \pi$ and $x = 3\pi$

Key concept: the minus sign in front of b swaps where the maximum and minimum occur.

M1 - $a + b = 7$ and $a - b = -1$ formed

A1 - $a = 3$

A1 - $b = 4$

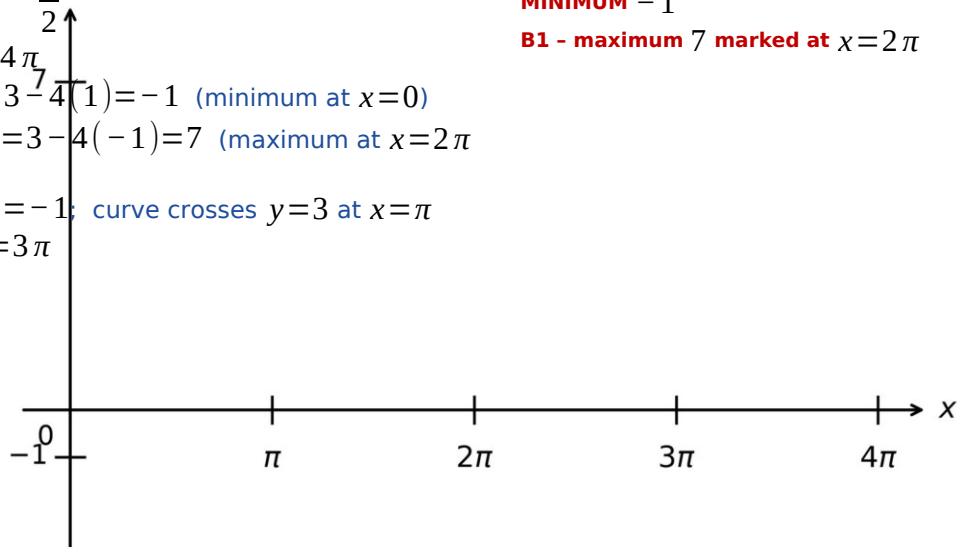
Trap - $a - b = 7$ comes from ignoring the minus

sign in $-b \cos \frac{x}{2}$.

B1 - period 4π

B1 - one complete cycle, starting and ending at the MINIMUM -1

B1 - maximum 7 marked at $x = 2\pi$



(c) Find the values of x , in terms of π , for which $f(x)=5$. [4]

$$3 - 4 \cos \frac{x}{2} = 5$$

$$-4 \cos \frac{x}{2} = 2$$

$$\cos \frac{x}{2} = -\frac{1}{2}$$

$$0 \leq x \leq 4\pi, \text{ so } 0 \leq \frac{x}{2} \leq 2\pi$$

basic angle = $\frac{\pi}{3}$; cosine is negative in
quadrants 2 and 3

$$\frac{x}{2} = \pi - \frac{\pi}{3} = \frac{2\pi}{3} \text{ or } \frac{x}{2} = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$$

$$\therefore x = \frac{4\pi}{3} \text{ or } x = \frac{8\pi}{3}$$

Key concept: transform the RANGE as well as the angle — $\frac{x}{2}$ runs to 2π only.

$$\text{M1 - } \cos \frac{x}{2} = -\frac{1}{2}$$

M1 - converts the range to $0 \leq \frac{x}{2} \leq 2\pi$

$$\text{A1 - } \frac{x}{2} = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\text{A1 - } x = \frac{4\pi}{3}, \frac{8\pi}{3}$$

Trap - working in the original range gives four answers, two of which are outside; doubling must come last.

----- End of Paper -----